

Background

X functions as a real-time reputation marketplace where public reactions surface within hours. Attention is fragmented across subcultures, sarcasm, and coordinated activity, so volume alone can misrepresent sentiment. Stakeholders need comparable snapshots showing where conversation is heading, not just how loud it is.

Problem Statement

Domain analysis tracks posts containing target keywords.

Co-domain analysis asks what the same users discuss beyond those keywords.

The key question: when does co-domain sampling improve targeting, message testing, and crisis prep, and when is it noise?

Objectives

1. Collect and preprocess X posts.
2. Score sentiment with a transformer model.
3. Compare domain vs. co-domain samples.
4. Extract themes using TF-IDF, LDA, and local LLM.
5. Identify positive vs. negative drivers of conversation.

Acknowledgements

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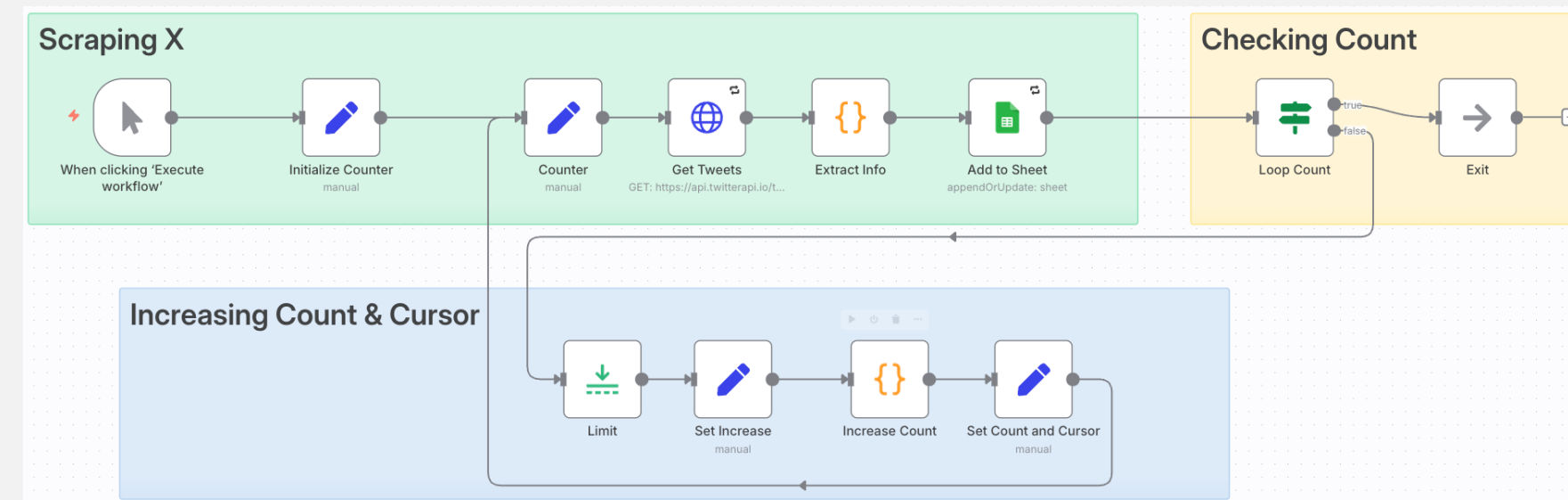


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Methods

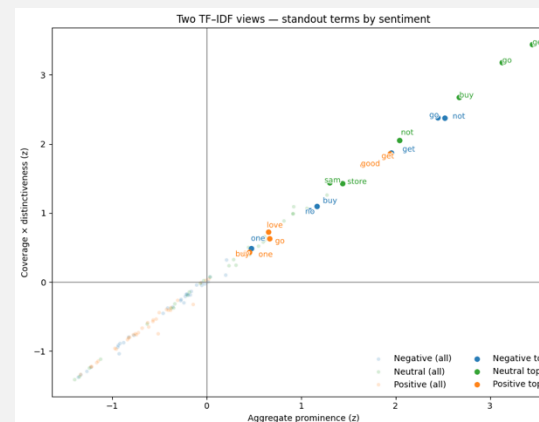
Data

- twitterapi.io
- *n8n*: node-based automation
- 60k posts across NFL, Keir Starmer, & Costco “Domains”
- 5-day post window



Key Terms

- Term Frequency-Inverse Document Frequency
- Compared Aggregate vs. weighted IDF: Spearman-rho of .999

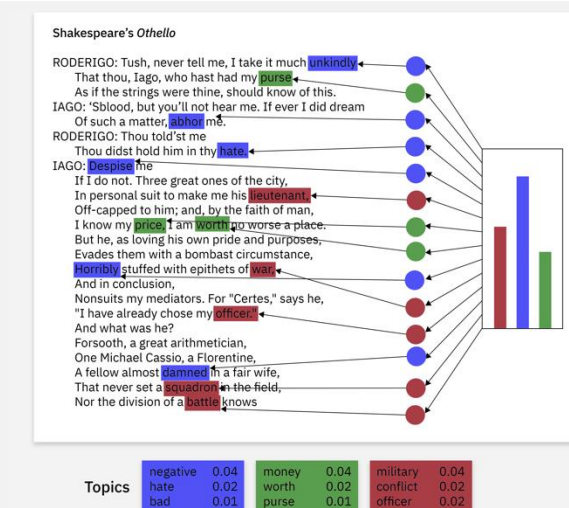


$$TFIDF_{t,d} = tf_{t,d} \times \log \left(\frac{N}{|\{d \in D : t \in d\}|} \right)$$

t Term
d Document
tf Frequency of term t in document d
N Total number of documents
D Number of documents d containing term t

Modeling

- Sentiment: *roBERTa* pretrained transformer
- Topics: Latent Dirichlet Allocation
- LLM: qwen3.5:9B-Q4 local via ollama



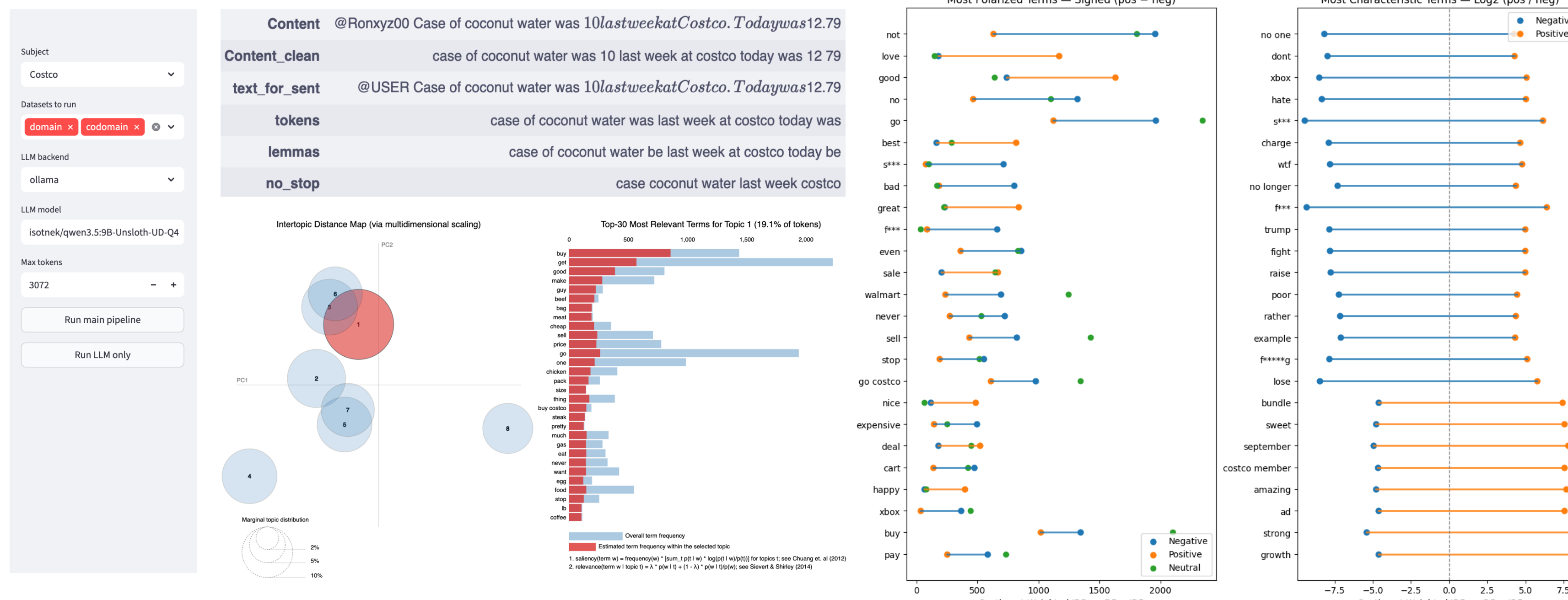
W Identity of word w in document d
Z Identity of topic of word w in document d
θ P(topic k occurring in document d)
φ P(word w occurring in topic k)
α Prior weight of topic k in a document
β Prior weight of word w in a topic

$$P(\mathbf{W}, \mathbf{Z}, \boldsymbol{\theta}, \boldsymbol{\varphi}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{i=1}^K P(\varphi_i; \boldsymbol{\beta}) \prod_{j=1}^M P(\theta_j; \boldsymbol{\alpha}) \prod_{t=1}^N P(Z_{j,t} | \theta_j) P(W_{j,t} | \varphi_{Z_{j,t}}),$$

Document Topic Topic Word Topic Sampling Word Sampling

Dirichlet Distributions Multinomial Distributions

Results - Domain Sampling: Costco



LLM Extracted Themes

Costco hot dog as a value symbol

- Hot dog price changes as shorthand for “is Costco still on our side?”.

Store choice & family/bulk tradeoffs

- Cross-shopping at Trader Joe’s, BJ’s, etc.
- Parents weighing bulk value, convenience, and “healthy enough” options for kids.

Price, protein, and quality concerns

- Repeated focus on price vs. quality/safety of staples (rotisserie chicken, meat, etc.).
- Reliability and transparency here act as trust levers.

Costco vs. Sam’s Club performance

- Direct comparisons on stock, electronics, and service.
- Perception is shaped by concrete outcomes (what’s on shelves, issue resolution).

Membership ROI, everyday essentials

- Joining/renewing justified by savings on batteries, tools, pantry staples, etc.
- Small frictions in staple items can tip sentiment on overall membership value.

Niche products, ethics, and lifestyle

- Interest in halal/healthier options, protein sources, specialty goods.
- Signals whether Costco aligns with specific dietary, cultural, or lifestyle needs.

Conclusions

Co-domain approach dilutes topic interpretability but contributes additional key word context for niche topics.

Raw frequency yields additional insights when paired with ranked IDF metrics.

References

Herk, N. (2025, March 14). How to Actually Scrape Twitter/X Data with n8n [Video]. YouTube. <https://www.youtube.com/watch?v=IEo7IAgj0UY>

Starmer, J. (2020, March 18). Latent Dirichlet Allocation [Video]. YouTube. <https://youtu.be/T05t-SqKArY?si=ZxP2DZt9vDicqIsc>